79.

LOCO out 26-6-24 22.00 hrs

SABARMATI DIESELS, LOCO CARE CENTRE, SABARMATI
Mechanical
IA/IB/IC Schedule of 3Phase Electric Locomotive [Inspection]

लोको संख्या Loco No 32458 क्लास Class IB शिड्यूलमें लेने का दिनांक Date taken under Sch 24/6/24

	आगमनIncoming / शिड्यूलSchedule		अंतिमनिरीक्षणFinal Inspection	
	अपर डेक Upper Deck	अंडरट्रकUnder Truck	अपर डेक Upper Deck	अंडर ट्रक Under Truck
Date/ Shift	24.6.24 GN	26/6/24 8/17		
Name of Technician	Manoj	Md Rafiq		
Designation/ Token No.	DM-2	MCF		
Signature				

Customer Feed Back/Bookings (Record feedback given by Loco pilot)

Sr. No.	Customer Feed Back/Bookings	Action Taken	Name of TCN / Attended By	Remarks, if any & Sign. of JE/SSE
1.	Machine room I and II both transformer silica gel colour change (pink)			
2.	CPA sefty valv blowing time 10.2 kg/cm²			
3.	79- valv lekege			
4.	Cab-1 LP wiper blade broken. ~~trade usage~~			
5.	CP-1 take maximum timing. 11 min 40 sec			
6.	CP-sefty valv blowing 9.5 KG (II CP)			
7.	CP-1 time taken ~~more~~. Less.			
	sefty valv			
8.	B.C pressure advance 3.8 (Cab-I/II)			
9.	machin room-I TMB II Flexible pipe broken.			
10.	Cab-I Water Jet not working.			
11.	58 Dz realy valv lekage.			
12.	MR 15 mintes में 2 kg drop होता है			

Inspection Wing Mechanical: Check and record Following

	CPA Loading Time (Pressure 0 to 8.0 kg/cm²)	Safety Valve Blowing	Pantograph-1 Raising	Pantograph-1 Lowering	Pantograph-2 Raising	Pantograph-2 Lowering	Compressor Loading Time (Pressure 8 to 9.5 kg/cm²) Comp-1	Comp-2	CP Safety Valve Blowing	Unloaded valve
Std. Value	8.0 Minutes	8.5± 0.25 kg/cm²	6 to 10 Second	Less than 15 Second	6 to 10 Second	Less than 15 Second	120 Second	120 Second	11.5± 0.35 kg/cm²	Operation time Approx. 12 Sec.
GI		10.2kg/cm²	7.1	8.9	6.9	10.9	120	95	9.5	8 sec
Final		8.7	9 SEC	11 SEC	7	10	110	81	>11	8 SEC

	MR safety Valve pressure Blowing	MR safety Valve pressure Closing	Duplex check valve setting/Brake Pipe Pressure by Auto Brake (A9) Cab-1	Cab-2	Feed Pipe Pressure Cab-1	Cab-2	Brake Cylinder Pressure by Direct Brake (SA9) Cab-1	Cab-2	Horn Sound Cab-1 LP	Cab-1 ALP	Cab-2 LP	Cab-2 ALP
Std. Value	10.5± 0.2 kg/cm²	9.5-9.8 kg/cm²	5.0 ± 0.1 kg/cm²		6.0 ± 0.2 kg/cm²		3.5±0.1 kg/cm²		Std. - Good			
GI	10.7	9.8	5.0kg	5.1kg	6.0kg	6.0kg	3.8kg	3.8kg	ok	ok	ok	ok
Final	10.7	9.8	5.0	5.0	6.0	6.0	3.5		Good	Good	Good	Good

	Computerized Control Brake (CCB) System if applicable: Passed successfully	Alerter and Penalty: Working	Air Flow: Working	Air Dryer: Working, Tower to change every minute (FTIL, SIL) & In every two minutes (KBIL)	Air Dryer: Humidity Indicator Colour Std-Blue	ADC: Working	MRC Time By both CP 0-10kg/cm²
Std. Val.	Passed successfully	Working	Working	Working, Tower to change every minute (FTIL, SIL) & In every two minutes (KBIL)	Std-Blue	Working	3.5 Minutes (Max.) For 1750 LPM or [8.5 Minute for each 1450 LPM CP]
GI			ok	TRIDENT	blue	wo	4 min 23 sec.
Final	E-70 Brake system		W.	working	BLUE	WORKING.	4 MIN 04 SEC.

FINAL-INSPECTION TIME BOTH CP I & II 0 TO 10 kg/cm² TIMING.

CP I :- 9 MIN 34 SEC

CP II :- 6 MIN 40 SEC

Signature of QAC JE/SSE

	CPA Pressure build up time 8.5 kg/cm^2	BP drop in 5 Minute	FP drop in 5 Minute	MR drop in 15 Minute	Press BPPB to apply or release parking brake. Check Pressure in Parking Brake gauge		VCB Pressure switch setting
Std. Value	60 Sec. Max.	Max. drop 0.15 kg/cm^2	Max. drop 0.70 kg/cm^2	Max. drop 1.0 kg/cm^2	6.0± 0.15 kg/cm^2		Opens 4.5± 0.15 kg/cm^2. Closes 5.5± 0.15 kg/cm^2
					Cab-1	Cab-2	
GI	59 Sec	no drop	No drop	drop	N/A		N/A
Final	58 SEC	NO DROP	NO DROP	0.8 DROP	Proried		proried

	ACP on 4mm test plate		Auto brake controller Position	BP (kg/cm^2)			BC (kg/cm^2)		
					Cab-1	Cab-2		Cab-1	Cab-2
Std Values	Working		Run	5.0 ± 0.10	5.0	5.0	0.0	0	0
			Initial Application	4.6 ± 0.10	4.6	4.6	0.40 ± 0.10	0.6	0.4
	Cab-1	Cab-2	Full Service	3.35 ± 0.20	3.4	3.4	2.50 ± 0.10	3.1	3.1
			Emergency	Below 0.30	0	0	2.50 ± 0.10	3.1	3.1
Gen. Insp.	OK	WORKING							
Final	WORKING	WORKING							

	Loco pilot cabin condition		Wiper Operation		Look out Glass		Fire Extinguisher	Sand box	Sander Operation		Cleaning of Loco
	Cab -1	Cab-2	Cab -1	Cab-2	Cab -1	Cab-2			Cab-1	Cab-2	
Std. Values	Good	Good	Working	Working	Clear	Clear	Sealed	Full	Working		Cleaned
Gen. Insp.	dirty	dirty	W	W	Dirty	dirty	Sealed		N/Working		Dirty
Final											

क्र.सं. SN	कार्यवाही कार्य /निरीक्षण का विवरण / (शिड्यूल) Detail of Work/ Inspection (Schedule)	कार्यवाही की गयी Action Taken	टीसीएन का नाम Name of TCN	हस्ताक्षर/ टिप्पणी Sign/ Remarks
	(A) Pneumatic System (When Loco is energised check, attend and record)			
1	Check correctness of reading of BP, FP, MR, BC, AFI & RS gauges.	ok		
2	Check on dummy of BP FP pipe.			
3	Check the working of PTDC	E-70 Brake system		
4	Check the proper functioning of all the gauges of the driver desk BP: (i) Normal (ii) with **7.5** mm hole BP should not drop below 4 kg/cm^2 in 60 second. FP: (i) Normal (ii) with **5.5** mm hole FP should not drop below 4.5 kg/cm^2 in 60 second. MR: (i) Mainline MR leakage without application of SA-9 Direct Brake (ii) Mainline MR leakage with application of SA-9 Direct Brake	BP- 4.5 FP-60	[illegible] 24.6.24	
5	(RS drop) Panto line air leakage 0.70 kg/cm^2 in 5 minutes.			
6	Check working of Panto selection switch.	Checked ok		
7	Position of isolating cock on pneumatic panel. (70,74,136 in open condition & 47 in close condition.)	Checked ok		
8	Drain moisture from MR and Inter Cooler.			
9	Check functioning of air dryer for purging, cycle time and change over with MR pressure of 8.0 kg/cm^2.	Checked ok		
10	Check Position of the Air dryer cock (13,14 in service & 187 in isolated)			
11	**Check air dryer charging when compressor is in working. (FTIL -120 sec/SIL-180 sec)**	Checked ok		
12	Check and attend leakage of air dryer, compressor valves, Pneumatic valves, Pneumatic panels & pipelines.	Checked ok		

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क्र.सं. SN	कार्यवाही कार्य /विवरण निरीक्षण का /(शिड्यूल) Detail of Work/ Inspection (Schedule)	कार्यवाही की गयी Action Taken	टीसीएन का नाम Name of TCN	हस्ताक्षर/ टिप्पणी Sign/ Remarks
B	BRAKE CONTROLLER (AUTO AND DIRECT)			
1	Check the functioning of driver's direct air brake & automatictrain brake.	Checked	Mantu 24.6.24	
2	Check the free movement of the brake controller.			
3	Perform repeated application and release of brakes with both auto and direct brake controller from both cabs and ensure that BP & BC pressure build up & drop is proper on the gauge. Ensure by physically going to individual main and parking brake cylinder that their operation is smooth and without air leakage.	Checked		
C	**CARBODY FILTER:**			
	Check condition and clean car body filter and refit.	Clean & Fit		Manoj Sharma 25/6/24
1	Thoroughly clean all roof and side body filters with water jet.	Checked		
2	Check tightness of Roof joint clamp and Roof top bolt.	Checked		
3	Remove Louvers& Fly screen and clean it. Repair if found defective.			
4	Remove the tread plate and clean the drainage channels.Check and adjust, if necessary, the door-operating rod (**IC Sch**)	Checked		
H	**LOOKOUT GLASS & SIDE WINDOWS**			
1	Clean any built up dirt and debris from the channel below the cab-sliding window. Clear the sliding window channel drains hole in the door opening.			
2	Clean look out glass with detergent water solution			
I	**WASHER / WIPER**			
1	Check the operation of windscreen wipers using the manual handles. Rectify any fault found.	Checked		
2	Check the condition of windscreen wiper blades. Replace if required.	Checked		
3	Check the wiper and parallel arm assembly for wear anddamage. Replace the arm or any part if there is excessive wear or damage. (IC Sch)	Checked		
4	Check the water level in the reservoir at No.1 and No.-2 end. Top up if required. Check wipers for proper operation	Checked	Mantu 24.6.24	
5	Clean the washer nozzle jet. (IC Sch)			
6	Check for damage of air and water pipes			
7	Examine all components for wear, looseness scoring andcracks, renew any defective components.	Checked		
8	Check seat-mounting points for integrity and ensure fixing or fully tightened.	Checked		
J	Key interlocking: Check all keys are present & the complete system is operational			
K	**Fire Extinguisher**			
1	Ensure availability of fire extinguishers and ensure that it is not expired.	Cab-1 30.12.24 Cab-2	Expiry date	
2	Check the condition of CO2 gas cylinder, its regulator and pressure gauge in both cabs. Check the CO2 gas pressures. Refill if necessary	Checked		Mantu 24.6.24
3	Clean pipeline by compressed air and check that there should not be any leakage.	-		
4	CO2 Gas pressure and date: Cab-1 82.0 kg/cm² 30.12.24 Cab-2 84.0 kg/cm² 30.12.24			

[signature]
Signature of QAG JE/SSE

क्र.सं. SN	कार्यवाही कार्य /निरीक्षण का विवरण / (शिड्यूल) Detail of Work/ Inspection (Schedule)	कार्यवाही की गयी Action Taken	टीसीएन का नाम Name of TCN	हस्ताक्षर/ टिप्पणी Sign/ Remarks
5	Clean fire detection unit pipeline air intake holes with soft non-metallic brush & vacuum cleaner.			
L	**OIL LEAKAGE IN MACHINE ROOM / CLEANING OF MACHINE ROOM.**			
1	Visually inspect the machine room for any signs oil leakage from traction converter, transformer, D.C. Link, and series resonant circuit capacitors and other electrical equipment and carefully clean the same. (**Caution : Electric supply room)**	checked ok		
2	Clean the complete machine room with the air pressurizedsupply and vacuum cleaner to remove all the dust particles.			
M	**SAND GEAR SYSTEM**			
	a) Check Sand box, cover gasket, Ejector and replace gasket if necessary.	All not working & adapter & flexible pipe missing		
	b) Check all sander pipe for alignment and check fasteners also.			
	c) Check and ensure tightness of Sand box foundation bolt.			
	d) Ensure sander pipe height 60mm above rail level.			
	e) Refill sand in all sand boxes.			
N	**OIL COOLING UNIT CASING WITH RADIATOR**			
1	Remove all dust, dirt and debris from the radiator chamber via the machine room access cover. (By using vacuum cleaner).	done		
2	Visually check the conservator tank along with its gauge glassfor any sign of crack / damage / oil leakage.	checked ok		
3	Measure air delivery before and after cleaning of radiator.			
4	Measure vibrations of oil cooling casing using vibration analyzer and take appropriate action	checked ok		
5	Visually check the casing and support arrangement for oilpumps on the casing for any sign of crack of the damage	checked ok		
6	Visually check the oil cooler radiator for any oil leakage from VPTFP and VPSR and externaldamage from top to bottom.	checked ok	manta 26.6.24	
7	**IC Sch: a)** Clean radiator with hot water jet and check that the radiator is not blocked.			
8	b) Tighten the front foundation bolts of casing mounted on to theradiator and check intactness of jail and Z-clamp welding crack.	checked ok		

Check and put (√) mark on the check box (CB) against each of the following items if the same has been checked. Write defects noticed and action taken in the Non-Schedule work table.

UPPER DECK [Inspect, Attend and Record]			
SN	**Description**	**CB**	**Remarks**
1.	Loose, missing and broken components/parts.	√	
2.	Clamping and condition of its welding/fastener.	√	
3.	Condition and working of all pressure gauges.	√	
4.	Working of all brake valves.	√	
5.	Leakage of air (MR)	√	
6.	Gauges, securing of loco pilot cabin.	√	
7.	Operation of Wipers.	√	
8.	Co-Action working of A9 Auto brake.	√	
9.	Fire Extinguishers (prescribed Nos.) 02/02 Big small	√	

Signature of QAG JE/SSE

10.	Check tightness of compressor foundation bolt & abnormal sound.	✓	
11.	check slackness of compressor fan and no vibration in comp.	✓	
12.	Check tightness of air dryer foundation bolt.	✓	
13.	Ensure Air dryer indicator blue colour	blue.	
14.	Ensure clamping of all pipe to avoid vibration and rubbing.		
15.	Ensure fitment of all split pins	✓	
16.	tightness of all bolts of Loco Roof Check	✓	
17.	Check window of both cab, shutter window door & corridor doors	✓	

Check and put (√) mark on the check box (CB) against each of the following items if the same has been checked. Write defects noticed and action taken in the Non-Schedule work table.

UNDER TRUCK [Inspect, Attend and Record]

क्र.सं. SN	विवरणDescription	CB	टिप्पणीRemarks
1.	Loose, missing and broken components/parts.	✓	
2.	Clamping and condition of its welding/fastener.	✓	
3.	Air leakage in bogie.	✓	M. Rafiq
4.	Overheated parts / Abnormal wear in parts.	✓	26/6/24
5.	Check and Ensure Brake block and Wheel clearance >10mm	✓	
6.			

(Remarks if any & sign):

Signature and Name of Technician for Inspection Incoming with date & shift: Mantu Kumar + Dinesh sharma
24.6.24

Signature and Name of JE/SSE for Inspection Incoming with date & shift:

Signature and Name of Technician for Inspection Final with date & shift:

Signature and Name of JE/SSE for Inspection Final with date & shift:

अन्य किसी भी जानकारी **Any Other Information:**

Signature of QAG SSE Incharge (Inspection)

Signature of QAG JE/SSE

SCHEDULE PROGRESS CHART

SN	Date	Shift	QC	Work Details	Name of JE/SSE	Sign of JE/SSE	Remarks
1	24/6/24	8/17		pre inspection.			
2	25/6/24	8/17		Carbody filter sandone.			
3	26/6/24	8/17		Comm sch & Transformer done			
4							
5							
6							
7							
8							
9							
10							
11							
12							
13							
14							
15							
16							
17							
18							
19							
20							
21							
22							
23							

Signature of QAG JE/SSE

SABARMATI DIESELS, LOCO CARE CENTRE, SABARMATI

IA/IB/IC SCHEDULE OF 3 PHASE ELECTRIC LOCOMOTIVE [UNDERFRAME]

पश्चिम रेलवे
Western Railway

लोको संख्या Loco No	32548	क्लास Class		शिड्यूल में लेने का दिनांक Date taken under Sch	

UNDERFRAME IA/IB/IC SCHEDULE & RECORDS

A-Check and record Hoot Wear, Flange Wear & Tread Wear (Record Wheel Diameter in Schedule or After TT)

Wheel No	Wheel Tread dia	Root Wear	Flange Wear	Tread Wear	Wheel No	Wheel Tread dia	Root Wear	Flange Wear	Tread Wear
1	1064.8	3.5	1.5	0.5	7	1049.5	4	3	1.5
2	1066.2	3.0	1.5	1.5	8	1052.2	2.5	1.5	0.5
3	1064.2	2.5	0.5	1.5	9.	1049.2	3	2	2.5
4	1065.8	1.5	1	1	10	1052.9	0.5	1	1
5	1064.5	2	1	1.5	11	1050.0	5	3	2.5
6	1066.0	2.5	1.5	0.5	12	1052.0	2	1.5	1

Avinash + Ramu
Only Dia + RF only

B- Check and Record in IC Sch Vertical &lateral clearance (Primary & Secondary) as per TC/0082 (REV-1) dated 17/09/2015

Standard Value	Primary vertical clearance (Cab-1 &2 Bogie) Actual Value Axle box & Bogie frame											
	1	2	3	4	5	6	7	8	9	10	11	12
27 to 35 mm												

Standard Value	Primary lateral clearance (Cab-1&2 Bogie) Actual Value Axle box & Bogie frame											
	1	2	3	4	5	6	7	8	9	10	11	12
15 to 22 mm												

NP

Standard Value	Secondary lateral clearance (Cab-1 Bogie) Actual Value Bogie frame & under frame			
	CAB-2 LP Side	CAB-1 ALP Side	CAB-2 LP Side	CAB-1 ALP Side
32 to 60 mm				

Standard Value	Secondary lateral clearance (Cab-1 Bogie) Actual Value Bogie frame & under frame			
	CAB-2 LP Side	CAB-1 ALP Side	CAB-2 LP Side	CAB-1 ALP Side
45 to 55 mm				

क्यूएजी जेई/एसएसई QAG-JE/SSE

PK

C- Check and record wheel gauge below Parameters: (IC Schedule) WHEEL GAUGE [Permissible limit 1596 mm, (159610.5 mm for RB/New loco)]

Axle No.	A	B	C	Average	Remarks
1					
2					
3					
4					
5					
6					

Check and record below Parameters in Schedule or if TT is done:

Axle No.	Standard (mm)		Cab-1		Cab-2	
Location	Max	Min	RI	LI	R2	L2
Buffer Length	6		640	638	642	640
Buffer Height			1080	1075	1080	1075
Rail guard height			110	106	105	106
CBC Height			1062		1055	

27/6/24

		Cab-1	Cab-2
STRIKER WEAR PLATE	New 8 mm, Condemn 2 mm		
SHANK WEAR PLATE	New 6 mm, Condemn 1 mm		

E- Measure D SHAKLE and write Coupling No.

CAB-1 CAB-2

26/6/24

Check and Record Gear case Oil Level:

TM	OIL FOUND	OIL TOP UP	TM	OIL FOUND	OIL TOP UP
1	5. Liter	1. Liter	4	5 Liter	1. Liter
2	5. Liter	1 Liter	5	5 Liter	1 Liter
3	Full	Full	6	5 Liter	1 Liter

	Hand Brake Operation	CB Coupler Operation & Condition		Transition Screw Coupling Operation & Condition		CBC Lock Pin		HP/FF Pipe	
Std. Value	Working	Normal		Normal		Intact		Intact	
		Cab-1	Cab-2	Cab-1	Cab-2	Cab-1	Cab-2	Cab-1	Cab-2
Gen. Insp.		ok	ok	ok	ok	ok	ok		
Final Insp.		ok	ok	ok	ok	ok	ok		

Brake Piston Travel :						Brake Shoe release	Standard 3/9/10 mm
Class of Loco	Standard value	Observed (mm)					
WAG7/WAP7	107 to 117 mm	Location-	1	2	3	4	
WAG9		Location-					

क्यूएजीजेई/एसएसई QAG-JE/SSE

Sr. No.	Detail of Work/ Inspection	Name Technician	Sign/Remark
F	**DRAFT GEAR AND COUPLERS**		
	1. Liberally apply general-purpose grease to the face of the centre buffer coupler at the end of the Locomotive		
	2. Check the intactness of the CBC bottom support plate bolts	Checked ok	
	3. Check the knuckles against crack by DPT (IC Sch)		
	4. Check knuckle by gauge no. 1&2 (130-135 mm by go no go gauge) (IC Sch)	Checked ok	
	5. Check knuckle nose wear by gauge no.3 go no go gauge (IC Sch)	Checked ok	
	6. Check intactness and functioning of CBC operating handle.		
	7. Check visually the striker casting of CBC assembly.	Checked ok	
	8. Check for working of CBC, Lock and unlock it, locking arrangement, working of TC	Checked ok	
	9. Check washer lock at knuckle pin bottom	Checked ok	26/6/22
	10. Check gap between Striker and Draft gear for draft gear slackness	Checked ok	
G	**BUFFERS**	Checked ok	
	1. Check buffers for any crack visually of the flange and intactness of foundation bolts, check nuts and split pins (4 nos. each),	Checked ok	
	2. Liberally apply general purpose grease to the face of the buffers at the ends of the Locomotive. Also smear grease on the buffer head plunger.	Checked ok	
	3. Conduct RDPT of Buffers to detect crack flaws. (IC Sch		
	4. Check and measure the buffer height and buffer length. (IC Sch)	Checked ok	
H	**SUSPENSION**		
	1. Examine the primary and secondary springs for signs of damage, cracks and broken end visually with torch light and tapping by a hammer & replace, if necessary	Checked ok	
	2. Examine different types of dampers for oil leakage and intactness of fasteners and end fittings	Checked ok	26/6/25
	3. Proper torque of all fasteners related to suspension arrangement and mounting of equipment in under frame (IC Sch)	Checked ok	
	4. Check the condition of various Spheris block in the primary and secondary suspension arrangement and of various equipment mounted in the under frame like axle guide link, traction motor and its support arm, gear case support arm for their healthiness (it should not be crack) and replace, if necessary.	Checked ok	
	5. RDPT of all TM nose and its support arm. (IC Sch)	NA	
I	**TRACTION BAR**		
	1. Check the well being and presence of the safety sling	Checked ok	
	2. Check the push/pull rod pivot heads for damage	Checked ok	26/6/25
	3. Examine the traction link assembly for signs of damage. Any defective item should be renewed	Checked ok	

क्यूएजी जेई/एसएसई QAG-JE/SSE

Sr. No.	Detail of Work/ Inspection	Name Technician	Sign/Remark
	4. Check the fastener at the traction ends and those retaining the traction rod for security. If a fastener is loose, it must be renewed and tightened to the specified torque. Ensure intactness of the locking plate. SM1-241		
	5. RDPT of traction bar at end (flange) portion & V-Ring	Checked	
	6. Housing flange. 6) RDPT of Pivot post for any crack (IC Sch)	Checked ok	
	7. Examine limit chain and links for wear, corrosion or damage	Checked ok	
	8. Inspect the safety slings, pins and R clips for wear and damage,	Checked ok	
	9. Check Aclathan/Vulkollon ring against any crack or shifting	Checked ok	
	10. Check gap in housing flange and traction link flange	Checked ok	
	11. Torque modified bolt M20 X65 by 260Nm (SMI 241)	Checked ok	
J	**BOGIE FRAME AND BRAKE GEAR**	Checked ok	
i	**BRAKE BLOCK**	Checked ok	
	1. Inspect the brake blocks for wear. If it is below the minimum limit, replace the complete locomotive set brake block.	Checked ok	
	2. Check the brake block key security	Checked ok	
	3. Check the brake rigging for loose, damaged or worn out parts Rectify as required.	Checked ok	
	4. Examine the bogie frame, transom beam & damper base for signs of crack/damage	Checked ok	
	5. Check fitment of Palm Pull rod J Bracket, hanger lever and TM safety rope wire.	Checked ok	
	6. Inspect the rubber bumps for damage deterioration. Replace if required	Checked ok	
	7. Examine all bogies mounted equipment (Including earthing straps) for signs of damage and security of fastening	Checked ok	
	8. Inspect the brake rigging rotation points and component contact surfaces. Rectify any faults found.	Checked ok	
	9. Inspect the bogie body safety pin eyehole at the bogie frame for wear or cracks (IC Sch)	NA	
	10. Maintain the vertical & lateral clearances (Primary & secondary) as per TC-082 (IC Sch)	NA	
	11. Check the earthing shunts provided on bogie and body.	Checked ok	
	Check intactness or axle box cover bolts **and sealing.**	Checked ok	
	12. Inspect brake gear for any worn out loose, damaged, hanging or dragging parts. Confirm intactness	Checked ok	
	13. Test for correct operation of the brake cylinder actuator unit.		
	14. Check the proper tightness of parking brake & service brake pneumatic pipes.	Checked ok	
	15. Check the external hitting marks & physical damage on the brake cylinder actuator unit.		
	16. Check the condition of parking brake cylinder bellows If damaged, replace them.	Checked ok	
	17. On parking brake cylinder pulling spindle & ring should be intact.		
	18. Check proper functioning of parking brake & service brake cylinder.	Checked	

क्यूएजी जेई/एसएसई QAG-JE/SSE

Sr. No.	Detail of Work/ Inspection	Name Technician	Sign/Remark
	19. Check the operation (manual releasing) of the parking brakes	NA	
	20. Check the bogie isolating cock, should be in proper position.		
	21. Check the wear of scrubber blocks, replace if required (WAP5only)	NA	
	22. Maintenance of Traction Motor support plate and bogie nose to Prevent crack/breakage of Traction Motor support plate (Holder for Traction motor suspension) should be done as per RDSO's SMI no. RDSO/2011/EL/SMI/0269 (Rev. '0') dtd, 18.05.11		
ii	**WHEEL SET AND AXLE BOX**		
	1. Examine all the wheels visually for following defects.		
	a) Cracks: if any, requires replacement.		
	b) Flats: Length 40mm or more requires re profiling.		
	c) Cavities: Isolated cavities of length 15mm or more or multiple cavities of length 10mm or more and less than 50mm apart require re-profile/replacement.		
	d) Metal builds up Height 1mm, or above and length 50mm or more requires re-profiling. For length less than 50mm., remove the build up with hand tools.	Nothie	
	e) Scoring and Grooving: Circumferentially around the wheel tread due to action of brake blocks require re-profile/replacement.		
	2. Measure and record the wheel diameters and tread profile. Refer RDSO's Tech. Circular no. ELRS/TC/0083 (Rev. "0"), (IC Sch)	NA	
	3. Disconnect the electrical cable from the earth return unit. Remove the earth return unit from the axle. Examine the bearings and rear sealing position for signs of grease leakage or deterioration. (IC Sch)	NA	
	4. Check the axle for any cracks with the help of ultrasonic flaw detector. At the same time visual inspection should be done. (IC Sch Every 06 month)	NA	24/6/24
	5. Grease Axle box after 300000 KM Engine Run	NA	
	6. Check the temperature of axle boxes. The axle boxes which are running warm or reported to have higher temperature in comparison with others shall be attended on priority by physically opening of axle box cover and outer angle ring. The suitable maintenance shall be done either in-situ or by dismantling the axle box.	OK	
iii	**GEAR CASE (WAP7/WAG9/WAG9H)**		
	1. Externally clean the gear case & its gauge glass. Inspect for any signs of oil leakage and any external hits/ damage. Verify integrity of gauge glass protective cover and clean oil sight glass for visibility.	Chandan	
	2. Check the oil level in the gauge glass and top up if required. Ensure intactness of the drain plug along with its sealing. Check its tightness.	Chandan	
	3. Check for any breakage of bolt and replace	Chandan	
	4. Check for intactness of all fasteners of gear case	Chandan	
	5. Tighten all gear case fasteners at prescribed torque	Chandan	
	6. Check the gear case oil visually and replace if found dirty or dust full level in the gear case and fill to the requirement	Chandan	
	7. Check Gear case oil for metal contamination	Chandan	
	8. Visually examine for fractures, the tyre profile for wear, flat, and skidding marks	Chandan	

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Sr. No.	Detail of Work/ Inspection	Name Technician	Sign/Remark
	9. Check wheel root, flange and Tread wear & record Change all unserviceable Brake Blocks	chem ok	
	10. Check condition and operation of Hand Brakes. Lubricate & adjust as necessary.	chem ok	
	11. Adjust Brakes and lubricate tunion		
	12. Lubricate Brake Gearing, Slack Adjusters, Buffers, CB Coupler Assembly	chem OK	
	13. Split the cotter pin, if new provided		
	14. Sch IC: a) Axle box old grease to remove and New grease to fill, old grease sample to be sent Lab for metal content check	chm ok	26/6/22
	b) Lubricate suspension roller tube bearing.		
	c) Check Transition screw coupling long and short shackle bar thickness by Go-No Go gauge (39.5 mm snap 1/25 mm snap 8)	chm ok	
	d) Carry out DFT of Buffer Foundation and DPT/MPT of CBC Knuckle clevis &Yoke. Check size of Clevis, Knuckle, coupling by Go-No Go Gauge as specified. As per SMI-213/MIG-76	chm ok	

Check and put (✓) mark on the check box (CB) against each of the following items if the same has been checked. Write defects noticed and action takes in the Non-Schedule work table.

UNDER TRUCK [Inspect, Attend and Record]

SN	Description	CB No.	Remarks
1.	Measure and record buffer & rail guard height.		
2.	If skidded, fill Wheel Sledding Complain Report Farm, FRMB/22 or if Biased wheel wear, fill up Check last for Based Wheel Wear Cause Invest, FILM/24)	chm ok	
3.	Proper Fitment of Brake Block and as Key		
4.	Rubbing of brake block with wheel in brake release position	chml ok	
5.	Movement of brake pull rod in brake applied position		
6.	Brake Shoe and Brake hanger Pin/Bolts	chml ok	26/6/22
7.	Brake piston movement.		
8.	Support arrangement for brake cross tie bars		
9.	CBC Drooping.	chml ok	
10.	Operation of both Transition screw couplings		
11.	Operation of CBC, Operating Rod and Coupler lock Pin	chml ok	
12.	Operation of CBC Knuckles		
13.	CBC retaining plate and support plate bolts		
14.	Side buffers and their mounting bolts.	chml ok	
15.	Bolts of Cattle guards and back supports.		
16.	Fitment of Rail Guard	chml ok	
17.	**<u>Gear case mounting bolts and 'C' clamp bolts</u>**		
18.	**<u>Suspension bearing cap bolts and it's sealing.</u>**	chml ok	
19.	**<u>Crack in TM Suspension taper roller bearing housing</u>**		

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SABARMATI DIESELS, LOCO CARE CENTRE, SABARMATI
IA/IB/IC SCHEDULE OF 3PHASE ELECTRIC LOCOMOTIVE
[COMPRESSOR &PNEUMATIC SYSTEM]

लोको संख्या Loco No 32458 | क्लास Class 1B | शिड्यूल में लेने का दिनांकDate taken under Sch 26/6/24

क्र.स. SN	कार्यवाही कार्य /निरीक्षण का विवरण / (शिड्यूल) Detail of Work/ Inspection (schedule)	कार्यवाही की Action Taken	टीसीएन का नाम Name of TCN	हस्ताक्षर/ टिप्पणी Sign/ Remarks
A	**मेन कंप्रेसर और बेबी कंप्रेसर Main Compressor and Baby Compressor**			
1	Check the pressure build up timing of each compressor independently in pre-inspection and take needful action during inspection. [From 0 to 10 kg/cm² i) With 1750LPM Comp-7 Minutes max. ii) With 1450LPM Comp- 8.5 Minutes max.]	cp-I – 11 minte 40 sec cp-II – 8 mine 03 sec.		
2	Feel the vibrations of the compressor in pre-inspection and taken needful action during inspection.	Checked		
3	Check for any external damage / abnormal sound / proper functioning of the compressor.	Checked		
4	Check the complete compressor support brackets for their healthiness.	Checked	manoj 24.6.24	
5	Visually examine the compressor mounting resilient blocks and take needful action.	Checked		
6	Visually examine the various interconnecting pipelines and its support clamps of the compressors for their healthiness / air leakage.	Checked		
7	Cleaning of input air filter and the primary oil filter.	Clean		
8	Clean the breather assembly	Clean		
9	Check and Drain the inter cooler / after cooler assembly and also observeits discharge and take needful action.			
10	Ensure healthiness / intactness of various protection guards of compressor.	Checked ok		
11	Check the condition of the coupling between compressor & motor	Checked		
12	Visually check the compressor fan blade for their healthiness	Checked		
13	Check the oil level in the compressor and top up as perrequirement.			
14	Carry out the inspection of each type of compressor in line with recommendation of respective manufacturer.	Checked		
15	Check compressor foundation base for crack (For under slung compressor)	Checked		28/6/24
B	**Aux. Compressor (CPA)**			
1	Clean the CPA externally. Clean Suction Air filter.	Clean		
2	Check oil level and condition of oil and top up if required with the oil specified (Servo Press 150).			
3	Check tightness of all nut bolts, foundation bolts.	Checked		
4	Measure and record time & pressure of the auxiliary compressor	Checked		
C	Sch IC: i) CP all discharge valves to remove.	—		
	ii) CP all discharge valves refit by overhauled	—		
	iii) CP oil to be change.	Changed		
	iv) Baby compressor/CPA oil to be change	Changed		
	CPA	Changed		
	CP-1	Changed		
	CP-2			
	v) Baby compressor/CPA safety valve (8.5 kg/cm²) to be replace by overhauled	—		

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क्र.सं. SN	कार्यवाही कार्य /निरीक्षण का विवरण / (शिड्यूल) Detail of Work/ Inspection (schedule)	कार्यवाही की Action Taken	टीसीएन का नाम Name of TCN	हस्ताक्षर/ टिप्पणी Sign/ Remarks
D	**AIR SUPPLY AND PNEUMATIC SYSTEM**			
1	NRV and unloader valves fail due to ingress of Carbon which extracted from CP. So. one cycle dismantling, checking and cleaning of NRV and unloader to be ensured in all type of 3-phase locomotives.			
E	**AIR DRYER**			
1	Inspect the air dryer for any damage and its mounting for any damage, crack in mounting plate, intactness of fasteners. Check for any hit mark on main reservoir and drain cock. Check for proper position of the air dryer angle cocks.	Checked ok		
2	Check the integrity of the welding of the hanging support.			
3	Check humidity indicators for blue colour. White indications mean the air dryer is not operating effectively. Desiccant shall be changed, if needed.	Checked ok		Mantu 24.6.24
4	Replace desiccant if found to be contaminated with either water or oil.	Checked ok		
5	Testing of air dryer for proper functioning to be done as perTechnical Circular no. **RDSO/2011/EL/TC/0108, Rev.'0' dtd. 7.3.11** for testing of air dryers and its efficacy on electriclocomotives.	Checked ok		
F	**E-70 BRAKE SYSTEM (PNEUMATIC PANEL)**			
1	Visually inspect the pneumatic panel noting any sign of air leakages and rectify accordingly.	Checked ok		
2	Clean the pneumatic panel externally.	Checked ok		
3	Check for correct position of the cut out isolating cock.			Mantu 24.6.24
4	Check all the cable connections for looseness or deficiency of cleats. If any inspection section is to be informed (**IC Sch**)	Checked ok		
5	Ensure provisioning and proper functioning of RS gauge, Switch, lamp & other gauges on pneumatic panel.	Checked ok		
6	Check the sealing of connectors, flexible pipe, regulator, air tank etc.			
G	**BRAKE ISOLATING COCKS:** Check for correct operations of BP & FP angle cocks. Check the condition of BP/FP pipe, cock & isolating handle for any hit markor crack. Check correct condition of Bogie Isolating cock.	Checked ok		
H	**BRAKE HOSES**: Examine the brake hoses at the ends of the locomotive to make sure that they are not chafed or damaged. Check couplingcocks for correct operation.	Checked ok		
I	**AIR RESERVOIR**			Mantu 24.6.24
1	Blow down and drain the reservoir until exhaust of complete water and oil. Check all drain cocks are closed on completion.	Checked ok		
2	Check for any hit mark on main reservoir & drain cock	Checked ok		
3	Ensure intactness / tightness of main reservoir foundation bolts.			
4	Check the rubber gasket at the inspection cover of air reservoir for any damage/leakage.	Checked ok		
5	Check the pressure regulator operates correctly			

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क्र.सं. SN	कार्यवाही कार्य /निरीक्षण का विवरण / (शिड्यूल) Detail of Work/ Inspection (schedule)	कार्यवाही की Action Taken	टीसीएन का नाम Name of TCN	हस्ताक्षर/ टिप्पणी Sign/ Remarks
J	**THROTTLE VALVE (IC Sch)**: Overhaul the throttle valve and change the rubber parts.			
K	**IN LINE FILTER (IC Sch)**: Remove the filter element and clean thoroughly.			
L	**HORN & HORN SWITCH**			
1	Check condition of horn handle for any damage.Check for proper operation of horn.	checked VM		
2	Check the condition of horn boots and replace if required	checked VK		
3	Remove the horn & horn switch and Overhaul. Replace thediaphragm of horn. (**IC Sch**)			
4	Check all angle cocks for proper operation	checked VM		
5	Ensure proper colour code of all the angle cocks.	checked VM		[illegible] 21/6/24
6	Check the tightness of the brake electronics control cards holding screws.			
7	Check all the pipes for leakage or damage. Check all the clamps for any damage and looseness.	checked VM checked VM		
M	वायुवीय प्रणाली **PNEUMATIC SYSTEM**			
1	MR Safety valve. ADC. Panto throttle valve (Only for conventional Pantograph) replace by Overhaul in **IC Sch.**			

Remarks if any

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SABARMATI DIESELS, LOCO CARE CENTRE, SABARMATI
CHECK SHEET FOR 3PHASE HR PANTOGRAPH ASSEMBLY (IA/IB/IC)

लोको संख्या Loco No 32458 क्लास Class WAG9HC शिड्यूल में लेने का दिनांक Date taken under Sch 26/6/24

SCHEDULE: IB PANTOGRAPH SR.NO. /Make//Type: .... LOCATION: ....

SN	Detail of Work/ Inspection (schedule)	Standard value	Action Taken PT-1	Action Taken PT-2	Remarks
1.	Ensure that wearing strips are without sharp edges and no gap between end strip and wearing strip.	Smooth and no gap	Smooth and no gap	Smooth & gap	
2.	Ensure availability of bolts. nuts. lock nuts. spring. washer. split pin etc.	All intact	All ok	All ok	
3.	Check the rubber air pipe for any leakage	No leakage			
4.	Check the rubber air bellow for any leakage/damage/rubbing	No leakage			
5.	Check lower frame assembly for any crack and play	No crack	No crack	No crack	
6.	Check upper frame assembly for any crack and breakage	No crack	No crack	No crack	
7.	Check coupling rod assembly for crack and its fitting intactness.	No crack & intact	No crack Intact	No crack Intact	
8.	Check diagonal rod for any crack both end side. (by DPT)	No crack	No crack	No crack	Amey Sharma 26.6.24
9.	Check plain bearing eye bush for freeness.	free	free	free	
10.	Check ball bearing friction of Base frame. Upper frame and Coupling rod.	ok	ok	ok	
11.	Check Cam wear on top of cable rope guide surface.	OK	ok	ok	
12.	Check condition of all shunts (not more than 5% strand broken)	OK	ok	ok	
13.	Check pantograph pan for any external hitting marks	No hitting marks	No hitting	No hitting	
14.	Verify all welded joint for cracks.	OK	ok	ok	
15.	Check the control box for any air leakage	No air leakage			
16.	Check the condition and operation of hydraulic damper.	Normal	Normal	Normal	
17.	Check Parallel guide bar for free function of Panto Pan Head	working	working	working	
18.	Ensure that the thickness of wearing strips above the condemning size	New – 39 mm Condemn – 26 mm	37.10 mm 36.07 mm	New fitted	
19.	Check the static contact force of pantograph at 6.4 kg/cm^2 MR pressure	70 N / 7Kg Load Test	ok Tight	ok Tight	
20.	Check the raising and lowering time of pantograph	6 to 15 second			
21.	Check the horizontality of panto pan with sprit level.	Level OK	ok	ok	
22.	Check the function of auto drop device (ADD)	Working	Doing	Doing	

SN	Detail of Work/ Inspection (schedule)	Standard value	Action Taken		Remarks
			PT-1	PT-2	
23.	Check the function of over reach detector (ORD)	**Working**	Dung	Dung	
24.	Check air filter for air leakage or not draining.	**No leakage**			
25.	Check the all pneumatic valve (3/2 valve, throttle valve and safety valve) for any leakage and proper function.	**No leakage**			
26.	Check pressure regulator for any leakage and proper function.	**No leakage**			
27.	Check the tightness of all bolts and thread locking compound to be provided in bearing base bolts.	**Provided**	[illegible]	[illegible]	
28.	Examine the roofline conductors for defects/looseness at joints.	**No defect**	No defect	No defect	
29.	Check the roof for any foreign material such as wire / clothpieces.	**No foreign material**	No foreign material	[illegible]	
30.	Check tightness, cleanliness and wear-tear of shunt connection between Panto and Roof line.	**Tight, clean**	Clean	Clean	Manoj Sharma 26-6-24
31.	Check that the pan head locking pins springs are in place and secure. Check that the head moves freely on its mounting pins attached to the apex frame. If movement is not free, check that the head is not twisted, the side beam are not bent/twisted & mounting pins are not loose or bent.	**OK**	ok	ok	
32.	Check tightness of the flexible shunt connecting screws. When a loose screw is found the shunt should be disconnected from the head &/or the pantograph frame, the mating surface cleaned & the shunt reconnected with new screws. Check all shunts for frayed strands and renew any shunt in which 5% or more of the strands are frayed.	**OK**	ok	ok	
33.	**IC Sch:** Clean filter element, if found brownish then replace with new.	**Clear**	—	—	

Remarks if any:

Signature and Name of Technician

Signature and Name of JE/SSE

58 - D2 Relay valv:

A.T.T. Reading - wheel dia

1	2	3	4	5	6
1051.1	1051.5	1052.2	1051.8	1052.0	1052.7
7	8	9	10	11	12
1037.0	1037.3	1038.9	1038.5	1036.4	1037.0

RFT Reading

	1	2	3	4	5	6	7	8	9	10	11	12
R	.5	1	1	1	1	1	.5	.5	.5	1	0	.5
F	0	.5	0	.5	0	0	.5	0	.5	.5	0	0
T	.5	.5	.5	1	.5	1	0	.5	.5	.5	.5	1

26/6/24

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SABARMATI DIESELS, LOCO CARE CENTRE, SABARMATI

IA/IB/IC Schedule of 3 Phase Electric Locomotive [Heavy Section]

लोको संख्या **Loco No** ________ क्लासClass ________ शिड्यूल में लेनेका दिनांक **Date taken under Sch** ________

Sr. No	Details of Work / Inspection (Schedule)	Action Taken	Name of TCN	Sign / Remarks
A	**V C B**			
1	Check insulator for crack. VCB roof bus bar, chips and flash marks. If damage in service, renew it			
2	Clean insulator, other parts that are dirty with soft, clean & dry cloth. (Do not use cleaning products based on fluoridated or chlorate compounds or sodium meta silicate), apply Silicon Spray.		manoj	
3	Check contact spring (Earthing contact)		Mahipal	
4	Check moving contact lever of auxiliary switch and tension spring & its mounting bolts for breakage			
5	Check the cables & lugs for broken cables, lugs looseness			
6	Check the main bushing condition, its connection and foundation bolts and clamps.			
7	Check the tightness of **HV** Connections (Torque 70Nm)			
8	**IC Sch:** a) Check for damage to connection of the earthing isolator, Cleaning & greasing, if necessary.			
	b) Check the torque of VCB fixing screw (Torque 70Nm)			
	c) Remove the bottom cover and check that the control air and electrical connection are tightened & Ensure that there is no leakage from pressure regulator			
B	**EARTHING SWITCH (HOM)**			
	a) Check the arm of earthing switch for any sign for any defect, play, crack etc. copper braid is undamaged and properly connected			
	b) Clean and lubricate the copper braid			
C	**PRESSURE SWITCH (IC Sch)**			
	Check the tightness of pneumatic connection			
D	**AUXILIARY SWITCH (IC Sch)**			
	a) Check drive screw for breakage or other damage			
	b) Check the parts of control rod and stay bolt for any deformation			
	c) Check the cables and lugs for broken cables, lugs looseness.			
	d) Securing of auxiliary blocks and support (Tightening torque-1.2Nm)			
	e) Check the working order of each auxiliary contacts			
E	**EARTHING CONNECTION.**			
	a) Check the earthing connection tightening torque 70Nm.			
	b) Connection of earthing system in 3-phase locomotives type WAP5, WAP7 & WAG9 (**As per RDSO/2007/EL/SMI/248 Rev.'0'** dated 22.11.2007)			

Signature of QAC JE/SSE

Sr. No.	Details of Work / Inspection (Schedule)				Action Taken	Name of TCN	Sign / Remarks
	c) Check Earthing shunt of Bogie to Body						
	Axle box 1-3		Axle box 2-4				
	Axle box 5-7		Axle box 6-8				
	d) Check Earthing shunt of Axle to Bogie						
	Axle box 2		Axle box 3				
	Axle box 6		Axle box 7				

Sr. No.	Details of Work / Inspection (Schedule)	Action Taken	Name of TCN	Sign / Remarks
F	MAIN TRANSFORMER			
1	Read off the oil level on the gauge situated on the conservator. Top up the oil as necessary and check for any signs of leakage. Clean the gauge with dry cloth	Checked		
2	Inspect the colour of the silica gel, if it is pink, replace with blue silica gel	Checked		
3	Examine the flanges of the pipe couplings and flexible hose that link the transformer and conservator	Checked	Naresh	
4	Check the availability of hosepipe over the oil pipe compensator and check stoochi coupling pipes for proper layout/fitment	Checked	Vikram	
5	Check the main Transformer and its protection cover for any damage, crack & oil leakage. Also refer instructions contained in **RDSO's letter no. EL/3.2.1/3-Ph / TC-0076 for arresting oil leakages cases"**	Checked		
6	Check the deformity of Transformer drain cock protection cover guard if deformed replace it	Checked		28/6/24
7	Perform the sample test on transformer oil. Check specific value of BDV, DGA, moisture and acidity and condition monitoring of loco transformer by Dissolve Gas Analysis. **(RDSO SMI 138 & OEM Doc. Dt. 27th Nov, 1995)**	Checked		
8	Check visually the condition of transformer foundation bolt and Nylon lock nuts for proper locking	Checked		
9	Check the earthing cable. Clean and visually inspect the insulator condition and Electrical connection of transformer and converter for crack & flashed mark and ensure red marking	Checked		
10	Visually inspect for leakage, damage, burning overheat signs and all fixing clamps condition of oil cooling metallic pipes, Prismatic level gauge, TFP bushing, HV bushing, Hose pipe etc.	Checked		

FINAL-INSPECTION TIME REPAIR

1. CP-I Loading time to much
2. BOTH SILICA GEL PINK.
3. CAB-1 B.C GAUGE LIGHT NOT GLOWING.
4. CAB-1 B.C PRESSURE EXCEED. (BOTH CAB)
5. CAB-1 LP SIDE WIPER WATER loading NOZZEL N.W.
6. All SANDING N.W.
7. CAB-2 BP & BC & AF2 LIGHT N.W.
8. MR-2 INSPECTION PLATE AIR LEAKAGE.
9. TMB-1 Sealing to do.
10. MR DROP in 15 MIN 2.8 kg/cm²
11. D-2 RELAY B.C AIR LEAKAGE
12. CP-2 DELIVERY PIPE RUBBING.

Signature of QA/G/JE/SSE